MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCMOS100 PC Hardware and Smartphone Troubleshooting

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	Which part of the PC stores the Operating System and user data? [a] RAM [b] CPU [c] HDD [d] GPU	U	1
2	Which slot type is used to install graphics card on a motherboard? [a] SATA [b] PCIe [c] USB [d] PCI	U	1
3	Which color wire in a SMPS connector typically represents +12V? [a] red [b] yellow [c] black [d] orange	K	2
4	What is the first step in a systematic troubleshooting methodology? [a] identifying the problem [b] implementing a solution [c] testing the solution [d] documenting the process	A	2
5	What is the common symptom of a faulty SMPS? [a] slow internet connection [b] no power to the device [c] frequent software crash [d] no power to the device	U	2
6	If you notice a loose connection on a PCB, what is the first step you should take? [a] replace the entire PCB [b] tighten the connection [c] Ignore it, as it's not important [d] reboot the device	A	2
7	Give the primary function of the Bluetooth unit in a cell phone? [a] to connect to wi-fi network [b] to process video signal [c] to enable wireless communication [d] to manage cellular network connections	U	2
8	Which component on a smart phone motherboard is primarily responsible for power management? [a] CPU [b] Battery connector [c] PMIC [d] Heat sink	K	3
9	Why is the GPS antenna often located near the top edge of a smart phone? [a] to be closer to the user [b] to minimize the interference [c] to reduce battery consumption [d] to enhance the phones aesthetic design	U	3
10	What is the primary function of boot mode configuration in a smartphone?	A	2

[a] manage phone contacts [b] trouble shoot and repair
[c] improve battery life [d] increase storage capacity

1 * 10 = 10

Part B

Short Answer Questions

Answer any **4** Questions

Each question carries 5 marks

11	Draw the block diagram of a PC and explain the working.	K	1
12	Explain the role of voltage regulators on a motherboard.	U	1
13	List the primary steps involved in a typical troubleshooting process in a PC.	K	2
14	Explain the role of the BIOS in a computer system. How does it contribute to the overall boot process?	U	2
15	List and explain the main sensors typically integrated into a smartphones motherboard?	U	2
16	Describe the different boot modes available in smart phones and their purposes.	U	2

4 * 5 = 20

Part C

Essay Questions

Answer any **2** Questions

Each question carries 10 marks

17	Discuss the architecture of the motherboard in a PC. What are the main features of each component?	U	1
18	Explain the working principle of a Switched Mode Power Supply unit in a PC with suitable diagram.	U	2
19	Explain the purpose of POST and its role in diagnosing hardware failure.	A	2
20	Explain the role of the RF unit in smartphones and discuss the types of antennas used in smartphones.	U	3

2 * 10 = 20